Health Effects of Noise

Noise Exposure

Noise is perhaps the most common occupational and environmental hazard. As many as 30 million Australians are exposed to potentially harmful sound levels in their workplaces. Outside of work, many people pursue recreational activities that can produce harmful noise. Non-occupational sources of noise include chain saws and other power tools, amplified music, and recreational vehicles such as motorcycles. Some types of toys for children can produce sounds capable of causing permanent hearing damage.

Common Sources of Noise

Sound	Loudness (dB)
Gunshot (peak level)	140 to 170
Jet takeoff	140
Rock concert, chain saw	110 to 120
Diesel locomotive, stereo headphones	110 to 120
Motorcycle, lawnmower	90
Conversation	60
Quiet room	50
Whisper	30 to 40

Noise can be described in terms of intensity (perceived as loudness) and frequency (perceived as pitch). Both the intensity and the duration of noise exposure determine the potential for damage to the hair cells of the inner ear. Even sounds perceived as "comfortably" loud can be harmful.

Sound intensity is measured as the sound pressure level in the decibel (dB) scale. Noise can cause permanent hearing loss at chronic exposures equal to an average of 85 dB or higher for an eight-hour period.

The decibel scale is a logarithmic scale. A 3-dB increase represents a doubling of the sound intensity. Four hours of noise exposure at 88 dB(A) is considered to provide the same noise "dose" as eight hours at 85 dB and a single gunshot, which is approximately 140 to 170 dB, has the same sound energy as 40 hours of 90-dB noise.

Health Effects

When sounds are measured in decibels, the softest sounds we can hear are about 1 dB and extremely loud sounds are about 150 dB. Most people find sounds louder than 130 dB very painful. To reduce the risks of hearing damage, workers in noisy jobs are now required to wear protective ear muffs. The cabins of noisy machinery are insulated to reduce noise. Apart from deafness, another common hearing problem caused by too much noise exposure is tinnitus. This is when you hear a constant ringing sound in your ears.